

ALPS: Bridging Academia-Industry Gap



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In this highly-competitive scenario, engineering firms prefer the most-skilled fresher candidate so as to reduce the time and resources spent on training. But, how do you provide industry based training in engineering and polytechnic colleges? Folks at iWave System Designs Pvt Ltd have come up with innovative and indigenous educational kits that solve this major issue to an extent.

Advanced learning platform and solutions (ALPS) is a product developed with the idea of unifying industrial processes for learning about and working on embedded systems. ALPS offers a generic architecture that lets the user easily integrate and evaluate several pre-defined peripherals like displays, sensors, communication and input/output (I/O) interfaces. This means that users get hands-on experience on industry peripherals, hence introducing them to the field of embedded systems, providing a rich visual and interactive experience, and enhancing their industry knowledge.

This interesting platform focuses on bringing about multidisciplinary industrial exposure in the educational segment. The entire kit was completely designed in India by the team led by Kumar Bhagavath, associate director, iWave, and it took a year to develop the final product out of just an idea.

Why ALPS

Today, most student projects are based on microcontrollers. These designs are targeted to

perform limited tasks, where the relationship of input and output are pre-defined. At times, a specific application may require features like image processing, real-time operations and touch-enabled LCD display. This calls for a processing capability that is many times faster, and high-speed memory devices. This is where ALPS comes handy. It is designed around a powerful processor that is scalable for automotive, industrial and consumer markets, and has a guaranteed lifetime of around ten to 15 years. Moreover, the product is integrated with various industry-relevant peripherals. Thus, ALPS provides an easy migration path for students to upgrade from microcontrollers to high-end processors.

The product

The hardware of this platform consists of two parts: a single-board computer (SBC) and a daughter board. SBC is a Pico-ITX board based on Freescale's i.MX6 ARM Cortex A9 processor, which is the intellectual property (IP) of iWave. It provides generic features such as processing, memory and power management with industry-relevant peripherals like Ethernet, USB, HDMI, audio, LAN and camera, among others. The daughter board is dedicated to student projects and is plugged to Pico ITX SBC, the specification of which can be changed based on educational needs.

The device also covers interfaces like sensors, motor drives, converters, switches and multiple display options that are typically being used by engineering students in their projects. These two boards along with a 17.8cm (7-inch) graphical LCD with touch screen forms the final product.

Having two separate boards for a single product gives manufacturers and users more adaptability. SBC is the brain of the device and has a complex architecture that should not be messed with. On the other hand, the daughter board is a simple PCB design that can be re-designed (add new



ALPS allows users to easily integrate and evaluate pre-defined peripherals such as displays, sensors and I/O interfaces